

TB Pitch Shifter

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Overview

Moves pitch and vocal character independently for natural shifts or dramatic transformations.

What this plug-in solves

Playback-speed changes tie pitch, duration, and perceived source size together. TB Pitch Shifter moves harmonic spacing across four octaves while formants can move independently, allowing octave changes, voice-size transformations, and designed tonal shifts without changing clip duration.

What you can expect

- Wide upward and downward pitch shifting
- Independent upward and downward formant shifting
- Automatable mono or stereo transformation
- Aligned bypass and repeatable session recall

How it works

TB Pitch Shifter rebuilds the source at a different pitch while allowing the formant, or perceived size of the voice and instrument body, to move separately. Keeping those two ideas independent makes both natural transposition and exaggerated character design possible.

Pitch chooses the musical movement; Formant changes character without requiring the same interval. Small independent changes can preserve a believable performance, while opposing or extreme directions create creature voices and synthetic doubles. Compare with the dry source and listen closely to attacks, consonants, and sustained notes.

Quick start

- 1 Start with both controls at zero.
- 2 Set Pitch for the required musical interval.
- 3 Adjust Formant to restore natural size or create a deliberate character.
- 4 Compare with aligned bypass.

- 5 Automate in musically meaningful transitions and leave headroom for reconstructed peaks.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Pitch

Moves the musical note and harmonic spacing in semitones. Positive values raise pitch; negative values lower it.

Formant

Moves perceived resonant size independently. Negative values generally sound larger/darker; positive values smaller/brighter.

Independent operation

Set one control to zero to change only the other. Link them manually only when a conventional speed-like size relationship is desired.

Programs

Init, Octave Up, Octave Down, Female to Male, and Male to Female provide direct starting points.

Latency and automation

Pitch analysis uses a fixed processing delay. Automation is smoothed, but very fast large jumps can still create an audible transition.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Formant	Moves the apparent vocal or instrument size without changing the played note. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Pitch	Moves the pitch of the processed sound up or down. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.

Practical uses

Natural octave double

- Set Pitch to +12 or -12.
- Move Formant slightly in the opposite direction if the source sounds too small or too large.
- Blend on a parallel track if the dry note must remain.

Expected result: A stable octave layer is created without changing clip duration.

Character transformation

- Leave Pitch near zero.
- Move Formant strongly positive or negative.
- Add downstream EQ only after the character is established.

Expected result: Perceived source size changes while the musical note stays centered.

Common pitfalls

Dense polyphony and transients may create spectral smearing. Enable only the repair section that matches the problem and use the smallest amount that makes the noise less distracting without removing useful detail.

Extreme shifts can expose phase or formant artifacts. Reduce width or movement and check the result in mono as well as stereo. Preserve the original spatial cues when they are important to the recording.

Fixed latency should be compensated by the host. Use the lighter quality setting while recording and reserve the heavier option for playback or export. Change latency-related modes while playback is stopped.